## Example 9.8

Calculate the seasonal index for the monthly sales of a product using the method of simple averages.

| Months | Year | | |
|---|---|---|---|
| | 2001 | 2002 | 2003 |
| Jan | 15 | 20 | 18 |
| Feb | 41 | 21 | 16 |
| Mar | 25 | 27 | 20 |
| Apr | 31 | 19 | 28 |
| May | 29 | 17 | 24 |
| June | 47 | 25 | 25 |
| July | 41 | 29 | 30 |
| Aug | 19 | 31 | 34 |
| Sep | 35 | 35 | 30 |
| Oct | 38 | 39 | 38 |
| Nov | 40 | 30 | 37 |
| Dec | 30 | 44 | 39 |

***Solution: Computation of seasonal Indices by method of simple averages.***

| Months | Year | | | Monthly Total | Monthly Averages |
|---|---|---|---|---|---|
| | 2001 | 2002 | 2003 | | |
| Jan | 15 | 20 | 18 | 53 | 17.67 |
| Feb | 41 | 21 | 16 | 78 | 26 |
| Mar | 25 | 27 | 20 | 72 | 24 |
| Apr | 31 | 19 | 28 | 78 | 26 |
| May | 29 | 17 | 24 | 70 | 23.33 |
| June | 47 | 25 | 25 | 97 | 32.33 |
| July | 41 | 29 | 30 | 100 | 33.33 |
| Aug | 19 | 31 | 34 | 84 | 28 |
| Sep | 35 | 35 | 30 | 100 | 33.33 |
| Oct | 38 | 39 | 38 | 115 | 38.33 |
| Nov | 40 | 30 | 37 | 107 | 35.67 |
| Dec | 30 | 44 | 39 | 113 | 37.67 |

Table 9.7

$$\text{S.I for Jan} = \frac{Monthly\ Average\ (for\ Jan)}{Grand\ average} \times 100$$

$$\text{Grand Average} = \frac{355.66}{12} = 29.64$$

$$\text{S.I for Jan} = \frac{17.67}{29.64} X100 = 59.62$$

$$\text{S.I for Feb} = \frac{26}{29.64} X100 = 87.72$$

Similarly other seasonal index values can be obtained.

| Months | Seasonal Index |
|---|---|
| Jan | 59.62 |
| Feb | 87.72 |
| Mar | 80.97 |
| Apr | 87.72 |
| May | 78.71 |
| June | 109.08 |
| July | 112.45 |
| Aug | 94.47 |
| Sep | 112.45 |
| Oct | 129.32 |
| Nov | 120.34 |
| Dec | 127.09 |

## Example 9.9

Calculate the seasonal index for the quarterly production of a product using the method of simple averages.

| Year | I Quarter | II Quarter | III Quarter | IV Quarter |
|---|---|---|---|---|
| 2005 | 255 | 351 | 425 | 400 |
| 2006 | 269 | 310 | 396 | 410 |
| 2007 | 291 | 332 | 358 | 395 |
| 2008 | 198 | 289 | 310 | 357 |
| 2009 | 200 | 290 | 331 | 359 |
| 2010 | 250 | 300 | 350 | 400 |

***Solution :***

Computation of Seasonal Index by the method of simple averages.



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